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Question

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Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item X , denoted by $r(X)$ and $w(X)$ respectively. Which one of them is conflict serializable ?

 $S_1 : r_1(X); r_2(X); w_1(X); r_3(X); w_2(X)$ $S_2 : r_2(X); r_1(X); w_2(X); r_3(X); w_1(X)$ $S_3 : r_3(X); r_2(X); r_1(X); w_2(X); w_1(X)$ $S_4 : r_2(X); w_2(X); r_3(X); r_1(X); w_1(X)$

This question was previously asked in

**UGC NET Computer Science (Paper 3)
Nov 2017 Official Paper**[Attempt Online](#)[View all UGC NET Papers >](#)1. S_1 2. S_2 3. S_3 4. S_4 **Answer** (Detailed Solution Below)

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Detailed Solution

CONCEPT

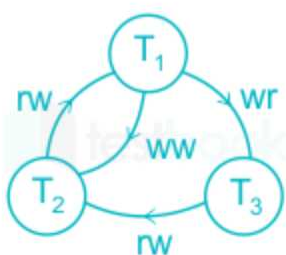
A schedule is called conflict serializable if it can be transformed into a serial schedule by swapping non-conflicting operations.

For a schedule to be conflict serializable, there should be no cycle present in its precedence graph.

EXPLANATION:

Option_1 – Not conflict serializable

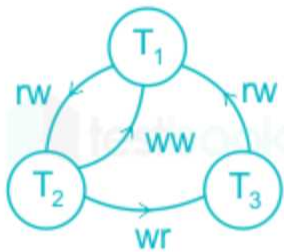
T ₁	T ₂	T ₃
r(X)		
	r(X)	
w(X)		
		r(X)
	w(X)	



- Cycle between T1 and T2.
- Cycle between T1 → T3 → T2

Option_2 – Not conflict serializable

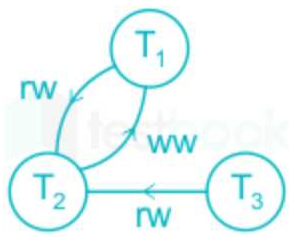
T_1	T_2	T_3
	$r(X)$	
$r(X)$		
	$w(X)$	
		$r(X)$
$w(X)$		



- Cycle between T_1 and T_2 .
- Cycle between $T_1 \rightarrow T_2 \rightarrow T_3$

Option-3 - Not conflict serializable

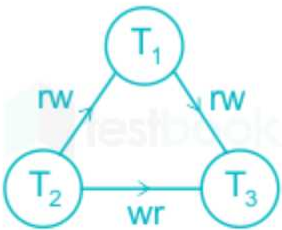
T_1	T_2	T_3
		$r(X)$
	$r(X)$	
$r(X)$		
	$w(X)$	
$w(X)$		



- Cycle between T1 and T2.

Option-4 - Conflict serializable

T ₁	T ₂	T ₃
	r(X)	
	w(X)	
		r(X)
r(X)		
w(X)		



- No cycle

Hence, Option_4 is a conflict serializable schedule